Tanmay Bhuskute

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Cincinnati, OH

SUMMARY

Backend Software Engineer with production experience building scalable SaaS systems, cloud infrastructure, and SQL-based data pipelines using Python, Node.js, AWS/GCP, Kubernetes, and IaC tools. Skilled at applying AI-driven automation and agentic tooling to accelerate delivery in fast-moving, resource-constrained teams.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Backend & Data: Node.js, Django, Flask, FastAPI, REST APIs, PySpark, Delta Lake, Data Pipelines.

Databases: PostgreSQL, MySQL, Data Warehousing.

Cloud & IaC: AWS, Google Cloud Platform, Azure, Terraform, Docker, Kubernetes, ArgoCD, Helm, GitLab CI/CD, Linux, SonarQube.

AI & Automation: Gemini, Vertex AI, LangChain, Agentic Tooling, PyTorch, scikit-learn, pandas.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, applying agentic AI tooling to cut manual evaluation effort.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**, applying infrastructure-as-code practices for consistent, scalable environments.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, SQL, PostgreSQL, FastAPI] Jul 2026

- Built ML data pipeline forecasting NYC taxi demand across **263 zones**, ingesting a **2.3M-row** dataset through a Spark-based bronze/silver/gold data warehouse; served predictions via FastAPI backed by PostgreSQL, cutting RMSE **43%** over baseline.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).